Senthamizhvelan M

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Professional Summary

AI & Data Science graduate specializing in agentic AI systems, retrieval-augmented generation, LLM tool orchestration, and workflow automation. Built and deployed a hand-rolled autonomous research agent and a multi-tool LangChain agent with hybrid search and text-to-SQL. Proficient in Python, SQL, FastAPI, Flask, and full-stack deployment, with experience owning AI systems from architecture through production.

Education

CK College of Engineering and Technology, Cuddalore *2026*
Bachelor of Technology in Artificial Intelligence & Data Science CGPA: 8.5/10

Anandhan Memorial Metric HSS Higher Secondary (12th): 87.53%
Anandhan Memorial Metric HSS Secondary (10th): 84.6%

Skills

Programming: Python, SQL, Java
AI/ML & Agentic Systems: RAG, LangChain, Agentic AI, Tool/Function Calling, Prompt Engineering, Hybrid Search (BM25 + Vector), Model Context Protocol (MCP)
Frameworks & APIs: Flask, FastAPI, REST APIs, HTML, CSS
Data & Databases: ChromaDB, MySQL, PostgreSQL, Pandas, NumPy
Tools & Platforms: Git, GitHub, Render, n8n, Data Structures & Algorithms

Professional Experience

CITPL (Cavin Info Tech Private Limited), AI/ML Intern *Feb 2026 – Aug 2026*

- Reduced manual workflow processing time by 20% by designing and deploying automated process pipelines with n8n across internal business systems.
- Built and deployed RAG-based knowledge-retrieval systems integrated via REST API endpoints for internal information access.
- Implemented Model Context Protocol (MCP) integrations connecting LLM workflows to internal tools and data sources, extending automation coverage beyond RPA-only pipelines.

Academic Projects

Autonomous AI Research Agent *Python, FastAPI, Dual-LLM Tool Calling*

- Architected a fully autonomous Plan → Act → Observe → Reflect research agent with a hand-built dual-format tool-call parser supporting JSON and XML LLM outputs.
- Engineered dual-engine resilience using Groq Llama 3.3 70B as the primary reasoning engine with automatic fallback to Gemini 2.0 Flash on errors or rate limits, capped at 15 reasoning iterations per query.
- Orchestrated web search, webpage reading, and calculator tools with prompt-enforced source verification requiring claims to be substantiated and conflicting information explicitly flagged.
- Built a real-time streaming UI using Server-Sent Events and deployed the application on Render.

NeuralRAG – Multi-Tool Agentic RAG Platform *Python, Flask, LangChain, ChromaDB, React/TypeScript*

- Built a LangChain tool-calling agent exposing six dynamically toggled tools: hybrid RAG search, text-to-SQL with SELECT-only safety guards, web search, system control, image generation, and chart generation.
- Architected hybrid retrieval combining dense ChromaDB and sparse BM25 retrieval through an ensemble retriever, with FlashRank reranking disabled in production after profiling exceeded the platform's 30-second request timeout.
- Implemented multi-provider LLM routing across Groq, Gemini, OpenRouter, and local LM Studio with automatic API-key-pool rotation to handle provider rate limits.
- Shipped a full-stack Flask and React/TypeScript application to Render, pre-downloading ONNX embedding weights to eliminate worker boot timeouts.